

Engineering Physics for EEE stream (BPHYE102/202)

QUESTION BANK

MODULE 3 –LASERS & OPTICAL FIBER

Laser

1. Give Acronym of Laser. Mention the properties of Laser.
2. Explain the terms induced absorption, spontaneous emission, stimulated emission.
3. Explain the requisites of a laser system.
4. Discuss the necessary conditions for laser action.
5. Obtain an expression for energy density of radiation under equilibrium condition in terms of Einstein's coefficients.
6. Explain the construction and working of CO₂ laser using energy level diagram.
7. Describe how a laser range finder is used in defence.
8. Describe the working of laser printer with diagram.

Optical Fibre

9. What is an Optical fibre? Explain how transmission of light takes place in optical fibers.
 10. Define the terms: i) Angle of acceptance ii) Numerical aperture iii) Fractional index change
 11. Obtain an expression for numerical aperture in terms of refractive indices of core and cladding of an optical fiber and then arrive at condition for propagation.
 12. With neat diagrams explain different types of optical fibers.
 13. What is attenuation? Explain the factors contributing to the fibre loss.
 14. Discuss point to point communication using optical fiber with the help of a block diagram.
 15. Describe the intensity based displacement optical fibre sensor (extrinsic sensor).
 16. Write a note on Attenuation Spectrum of an Optical Fiber with Optical Windows.
 17. Discuss the advantages and disadvantages of optical communication system over other system.
-

